Data Dictionary

A readable reference to the fields in the cleaned temperature and rainfall datasets. Observed ranges are the minimum and maximum values present in the supplied data dictionary.

Temperature dataset

Field	Data type	Unit	Source / derivation	Allowed range	Observed range
city	object	-	added by us	Kolkata, Thiruvananthapuram, Visakhapatnam	—
date	datetime64[ns]	date	calendar rebuilt by us	2022-06-01 to 2025-08-31	2022-06-01 00:00:00 to 2025-08-24 00:00:00
year	int32	year	derived from date	2022 to 2025	2022 to 2025
month	int32	month number	derived from date	1 to 12	1 to 12
TMAX	float64	deg C	NOAA GHCN-Daily (tenths / 10)	15 to 50	20.8 to 44.6
fc_tmax_lead1	float64	deg C	Open-Meteo Previous Runs, gfs_seamless, highest hour of the day	15 to 50	20.7 to 46.2
fc_tmax_lead2	float64	deg C	Open-Meteo Previous Runs, gfs_seamless, highest hour of the day	15 to 50	21.7 to 46.5
fc_tmax_lead3	float64	deg C	Open-Meteo Previous Runs, gfs_seamless, highest hour of the day	15 to 50	22.4 to 47.2
fc_tmax_lead4	float64	deg C	Open-Meteo Previous Runs, gfs_seamless, highest hour of the day	15 to 50	21.3 to 47.0
fc_tmax_lead5	float64	deg C	Open-Meteo Previous Runs, gfs_seamless, highest hour of the day	15 to 50	22.6 to 47.1
fc_tmax_lead6	float64	deg C	Open-Meteo Previous Runs, gfs_seamless, highest hour of the day	15 to 50	20.7 to 47.0
fc_tmax_lead7	float64	deg C	Open-Meteo Previous Runs, gfs_seamless, highest hour of the day	15 to 50	20.8 to 47.7

Rain dataset

Field	Data type	Unit	Source / derivation	Allowed range	Observed range
city	object	-	added by us	Kolkata, Thiruvananthapuram, Visakhapatnam	—
date	datetime64[ns]	date	calendar rebuilt by us	2024-01-20 to 2025-08-31	2024-01-27 00:00:00 to 2025-08-24 00:00:00

Field	Data type	Unit	Source / derivation	Allowed range	Observed range
year	int32	year	derived from date	2024 to 2025	2024 to 2025
month	int32	month number	derived from date	1 to 12	1 to 12
PRCP	float64	mm per day (08:30 to 08:30 IST)	NOAA GHCN-Daily (tenths / 10)	0 to 500	0.0 to 132.1
obs_status	object	-	derived from PRCP	recorded_zero, recorded_wet	—
fc_rain_lead1	float64	mm per day (09:00 to 08:00 stamps)	Open-Meteo Previous Runs, gfs_seamless, sum of 24 hours	0 to 500	0.0 to 165.7
fc_rain_lead2	float64	mm per day (09:00 to 08:00 stamps)	Open-Meteo Previous Runs, gfs_seamless, sum of 24 hours	0 to 500	0.0 to 237.4
fc_rain_lead3	float64	mm per day (09:00 to 08:00 stamps)	Open-Meteo Previous Runs, gfs_seamless, sum of 24 hours	0 to 500	0.0 to 217.3
fc_rain_lead4	float64	mm per day (09:00 to 08:00 stamps)	Open-Meteo Previous Runs, gfs_seamless, sum of 24 hours	0 to 500	0.0 to 249.4
fc_rain_lead5	float64	mm per day (09:00 to 08:00 stamps)	Open-Meteo Previous Runs, gfs_seamless, sum of 24 hours	0 to 500	0.0 to 71.1
fc_rain_lead6	float64	mm per day (09:00 to 08:00 stamps)	Open-Meteo Previous Runs, gfs_seamless, sum of 24 hours	0 to 500	0.0 to 66.6
fc_rain_lead7	float64	mm per day (09:00 to 08:00 stamps)	Open-Meteo Previous Runs, gfs_seamless, sum of 24 hours	0 to 500	0.0 to 108.9